

CHEMICAL BONDING: MASTER TEXTBOOK

1. Introduction

Why do atoms bond?

Atoms bond to achieve stable electronic configurations (usually the octet rule). Bonds form due to attractive electrostatic forces between atoms.

2. Primary Bonding Types

- **Ionic Bond:** Complete transfer of electrons from metal to non-metal (electrostatic attraction).
- **Covalent Bond:** Sharing of electron pairs between atoms (often non-metals).
- **Metallic Bond:** Sea of delocalized electrons holding metal cations together.

3. Molecular Geometry and Hybridization

VSEPR Theory

Valence Shell Electron Pair Repulsion theory predicts geometry based on the minimization of repulsion between electron pairs.

Hybridization

Mixing of atomic orbitals to form new hybrid orbitals (sp , sp^2 , sp^3). This explains the observed bond angles and molecular shapes.

4. Molecular Orbital Theory (MOT)

Bonding vs. Antibonding

MOT describes electrons in molecules occupying molecular orbitals. **Bonding orbitals** stabilize the molecule, while **Antibonding orbitals** destabilize it.